COMP 018 – DATABASE ADMINISTRATION

ACTIVITY 6: DATABASE BACKUP AND RECOVERY

Progress Report – Group 2

Evaluating Group: Group 2
Project Evaluated: CampusSafe – Campus Hazard & Safety Reporting System
Project Owner: Group 7
Database: grp07_campussafedb
DBMS: PostgreSQL 14
Backup Date: November 5, 2026

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1. Introduction

This activity evaluates the backup and recovery procedures of Group 7's CampusSafe database. The purpose is to demonstrate that the database can be backed up, inspected, restored, and recovered while preserving its records, constraints, and access-control configuration.

All restoration and recovery procedures were performed using separate test databases. The original CampusSafe database was not intentionally modified during these tests.

2. Backup Plan

A PostgreSQL logical backup using custom archive format (-Fc) was selected.

Backup strategy:

- Backup method: PostgreSQL pg_dump
- Format: Custom archive
- Storage: Protected local folder outside the Git repository
- Verification: File-size check, SHA-256 checksum, and archive inspection

- Proposed schedule: Daily backups
- Proposed retention: Weekly backups retained for one month and monthly backups retained for six months
- Planned Recovery Point Objective (RPO): 24 hours
- Target Recovery Time Objective (RTO): 4 hours
- The Activity 4 roles are cluster-wide objects and are not included in a single-database backup. Their definitions and memberships must therefore be managed separately.

3. Backup Creation and Inspection

The backup was created using PostgreSQL's pg_dump utility.

Backup information:

Item	Result
Database	grp07_campussafedb
Backup filename	grp07_campussafedb_activity6.dump
Backup format	Custom (-Fc)
File size	47,160 bytes
Backup account	campussafe_backup
Date	November 5, 2026

SHA-256 checksum:

E6D5515FC5734BC9BEBDBF66EDAC103DA770004C1994B9E9
7F71D05E83B3EC65

The archive was inspected using pg_restore --list. The inspection identified 118 table-of-contents entries, including nine tables, table data entries, sequences, indexes, constraints, and access-control entries.

The backup file was stored locally and was not committed to GitHub.

4. Restoration and Verification

The backup was restored into a separate database named:

grp07_campussafedb_restore_test

The restoration command completed without reported errors.

4.1 Table Verification

All nine CampusSafe tables were present after restoration:

1. app_user
2. assignment
3. attachment
4. campus_location
5. hazard_category
6. hazard_report
7. maintenance_action
8. report_status_history
9. role

4.2 Record Verification

Table	Restored records
app_user	6
hazard_report	16
report_status_history	60
role	5

4.3 Constraint Verification

Constraint type	Count
CHECK	7
Foreign key	14
Primary key	9
Unique	5

These results indicate that the tested records and database integrity constraints were preserved.

5. Activity 4 Access-Control Verification

The six CampusSafe roles were found in the PostgreSQL cluster.

Role	Login allowed	Superuser

app_runtime_grp	No	No
app_readonly_grp	No	No
app_backup_grp	No	No
campussafe_app	Yes	No
campussafe_report	Yes	No
campussafe_backup	Yes	No

The login-to-group memberships were also verified:

- campussafe_app → app_runtime_grp
- campussafe_report → app_readonly_grp
- campussafe_backup → app_backup_grp
- The restored database showed 45 table-permission entries across the three group roles.

Access Tests

Allowed operation: The campussafe_report account successfully connected to the restored database and retrieved five hazard reports using a SELECT query.

Denied operation: An UPDATE attempt by campussafe_report returned:

ERROR: permission denied for table hazard_report

This confirms that the reporting account could read the tested reports but could not modify them.

6. Recovery Simulation

A second separate database was created:

grp07_campussafedb_recovery_test

The same backup file was restored into this database using pg_restore. The operation completed without reported errors.

The recovered records were verified:

Table	Recovered records
app_user	6
hazard_report	16
report_status_history	60
role	5

The counts matched those from the earlier restoration test.

This was a controlled recovery test using a fresh database. It demonstrated that the backup could rebuild the tested database contents without changing the original database. It was not an actual production-failure simulation.

7. Evaluation and Recommendations

The backup and recovery tests demonstrated that the CampusSafe logical backup was readable and could be restored into separate databases. The tested tables, selected records, constraints, and Activity 4 permissions were present after restoration.

Recommendations:

1. Perform regular database backups according to the proposed schedule.
2. Keep backup files in protected storage outside the source-code repository.
3. Verify backup files through archive inspection and periodic restoration tests.
4. Maintain a separate procedure for backing up and recreating cluster-wide roles.
5. Protect backup credentials and restrict access to backup files.
6. Record backup and restoration results, including errors and actual recovery times.

8. Limitations

- Only the Group 7 CampusSafe project was evaluated in this progress report.
- The recovery exercise used a separate test database rather than an actual production failure.
- The four-hour RTO is a target and was not independently validated through a timed recovery test.
- The 24-hour RPO is a planned objective, not a measured result.
- Only selected table record counts were checked in the verification queries.

9. Feedback to Project Owner

The backup and restoration tests showed that the database could be backed up and restored into separate test databases. The tested access-control configuration was also present after restoration.

Feedback status: To be completed after communicating the evaluation results to Group 7.

10. AI Usage Declaration

AI assistance was used to help organize the backup and recovery documentation, prepare SQL commands, explain PostgreSQL procedures, and draft sections of this progress report.

The group executed the database commands, inspected the backup archive, performed the restoration and recovery tests, and checked the reported results.

11. Conclusion

The Group 2 evaluation of Group 7's CampusSafe database successfully demonstrated logical backup creation, archive inspection, restoration, access-control verification, and a separate recovery test.

The tested restoration and recovery procedures completed without reported errors. The results provide evidence that the created backup can be used to restore the tested CampusSafe database contents while preserving the verified access-control configuration.